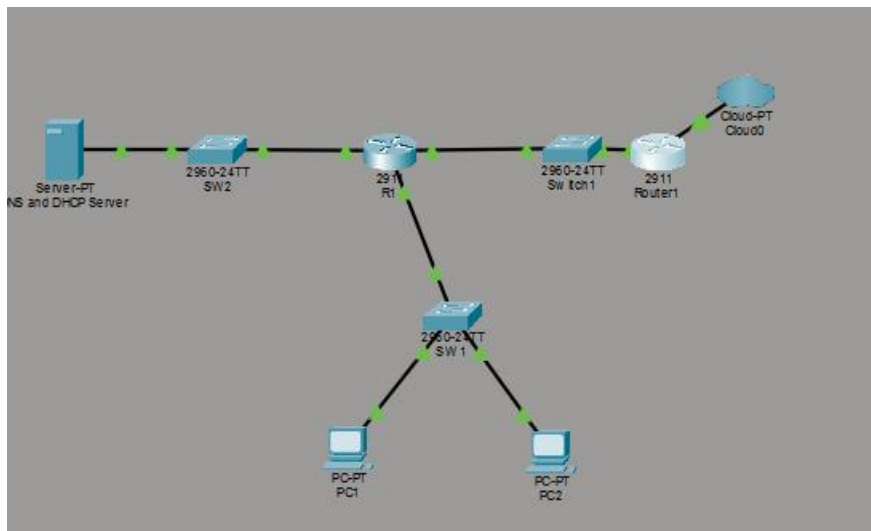
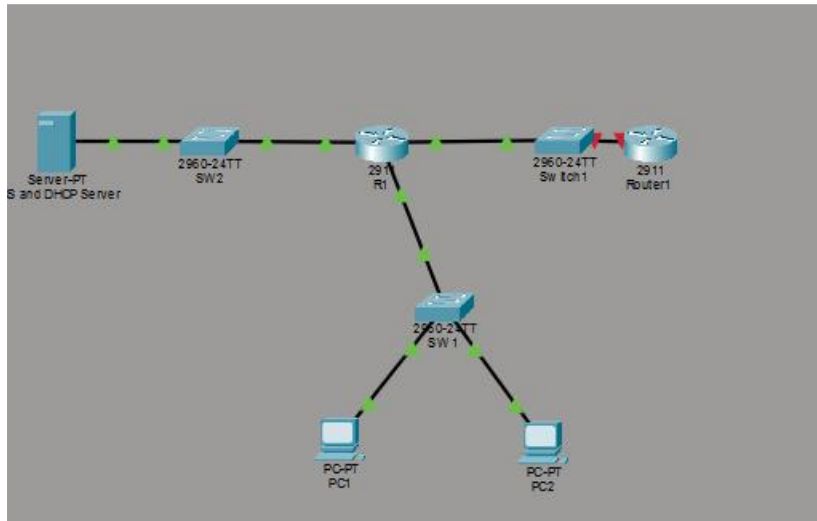


Topology



```
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface      IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0    unassigned      YES unset    up                  up
GigabitEthernet0/1    unassigned      YES unset    administratively down down
GigabitEthernet0/2    unassigned      YES unset    up                  up
Serial0/3/0         unassigned      YES unset    administratively down down
Serial0/3/1         unassigned      YES unset    administratively down down
Vlan1             unassigned      YES unset    administratively down down
Router#
```

Router 1

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.100.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#ip dhcp excluded-address 192.168.100.1 192.168.100.10
Router(config)#ip dhcp pool R1-OUTSIDE
Router(dhcp-config)#network 192.168.100.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.100.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
```

```
Router#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0 unassigned      YES DHCP    up              up
GigabitEthernet0/1 10.10.20.1      YES manual  up              up
GigabitEthernet0/2 unassigned      YES unset   up              up
Vlan1               unassigned      YES unset   administratively down down
Router#
```

DHCP

```
Router#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0 192.168.100.11  YES manual  up              up
GigabitEthernet0/1 unassigned      YES unset   administratively down down
GigabitEthernet0/2 unassigned      YES unset   up              up
Serial0/3/0         unassigned      YES unset   administratively down down
Serial0/3/1         unassigned      YES unset   administratively down down
Vlan1               unassigned      YES unset   administratively down down
Router#
```

```
Router#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0 unassigned      YES DHCP    up              up
GigabitEthernet0/1 10.10.20.1      YES manual  up              up
GigabitEthernet0/2 unassigned      YES unset   up              up
Vlan1               unassigned      YES unset   administratively down down
Router#
```

```
Router(config)#ip dhcp pool CAMPUS-LAN
Router(dhcp-config)#network 10.10.10.0 255.255.255.0
Router(dhcp-config)#default-router 10.10.10.1
Router(dhcp-config)#dns-server 10.10.20.10
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 10.10.10.1 10.10.10.5
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip dhcp binding
IP address          Client-ID/       Lease expiration   Type
                   Hardware address
Router#
```

PC1

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

DHCP request failed.

IPv4 Address

Subnet Mask

Default Gateway

0.0.0.0

DNS Server

0.0.0.0